

Beyond Linear Reasoning: Combining Chain-of-Thought with A* Search for Robust Multi-Strategy Inference

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Abstract

Chain-of-thought (CoT) prompting elicits useful reasoning from large language models, but its strictly linear generation can be brittle on problems that require backtracking, multi-hop synthesis, or comparison across competing hypotheses. We study a complementary alternative: a budgeted A* search over reasoning traces, where an LLM proposes successor steps and a conservative heuristic prioritizes which partial traces to expand. Theoretically, we prove that the proposed A* search is admissible under mild conditions. Across StrategyQA, LogicQA, and HotpotQA, we find strong instance-level complementarity between CoT and A*: the oracle that selects the correct strategy per example substantially outperforms either method alone. We then evaluate routing strategies based on semantic entropy, a feature-based random forest, and LLM-based pairwise adjudication. Routing improves accuracy, but still recovers only part of the oracle gain, showing that the main challenge is not search itself but deciding when to use it. Qualitative and human analysis further show that A* helps most on non-linear evidence aggregation, hypothesis elimination, and multi-hop chaining, whereas CoT remains stronger on direct factual recall and globally coherent reasoning. Overall, our results position A* not as a replacement for CoT, but as a complementary reasoning strategy whose value depends on the structure of the problem. Code is available at: <https://anonymous.4open.science/r/astar-reasoning-EC39>

1 Introduction

Chain-of-thought (CoT) prompting (Wei et al., 2022) has become the standard way to elicit step-by-step reasoning from large language models (LLMs). By generating intermediate reasoning steps before a final answer, CoT often improves performance on arithmetic, commonsense, and symbolic tasks. However, its core generation process

is still strictly *linear*: the model generates a single trace $s_1 \rightarrow s_2 \rightarrow \dots \rightarrow s_T$, and later steps are conditioned on all earlier generated contents. This design creates three potential problems. First, an early mistake can propagate through the entire trace because the model has no built-in backtracking mechanism. Second, questions that require integrating several independent pieces of evidence can be difficult to solve with a single left-to-right narrative. Third, CoT has no explicit search objective that trades off trace length and alternative hypotheses.

These limitations motivate search-based reasoning. If the model can explore multiple partial traces, then it can reconsider early commitments, compare candidate branches, and prioritize promising reasoning states. In this paper, we study a budgeted A*-search procedure over natural-language reasoning traces. The language model generates successor steps, a step cost penalizes both trace length and low-confidence steps, and a heuristic prioritizes which partial trace to expand next. We pair this search procedure with standard CoT and ask a deliberately narrower question than “does search replace CoT”? Our focus is whether search and linear reasoning are *complementary*, and whether simple routing mechanisms can exploit that complementarity.

Motivating example. On StrategyQA, linear CoT can pursue a locally plausible but ultimately irrelevant chain, while A* can surface the decisive evidence earlier by exploring alternative partial traces. For example (§Figure 1), on the question “Is shrimp scampi definitely free of plastic?”, CoT misses the key fact of microplastic accumulation in seafood, whereas A* reaches that premise early and answers correctly. This illustrates the central hypothesis of the paper: CoT and search are complementary, and the real challenge is deciding when to use which.

This pattern persists at scale. Across all three

benchmarks, CoT and A* solve different subsets of examples. On StrategyQA, A* outperforms CoT (74.80% vs. 64.52%), yet the oracle reaches 84.59%. On LogicQA, CoT is slightly stronger (45.78% vs. 44.85%), but the oracle still reaches 59.29%. On HotpotQA, A* again leads (80.10% vs. 76.80%), while the oracle reaches 94.20%. Together, these gaps show that the main opportunity is *strategy selection*: different questions favor different reasoning algorithms.

Contributions. We make three contributions. (1) We formalize a *budgeted* A* reasoning procedure for LLMs, define a conservative heuristic, and state explicit assumptions under which the heuristic is admissible for the underlying path-cost objective. Over standard search based objective, proposed A* search is theoretically superior because of its admissibility property. (2) We provide an empirical analysis across StrategyQA, LogicQA, and HotpotQA datasets, showing substantial complementarity between linear CoT and search-based reasoning. (3) We evaluate routing strategies: semantic entropy, a feature-based random forest, and an LLM-as-judge and critic—and pair the quantitative results with a diagnostic qualitative taxonomy of when each reasoning strategy helps or fails.

2 Methodology

We first formalize CoT and search-based reasoning under a common notation and then describe the routing mechanisms that choose between them.

2.1 Notation and Problem Setup

Let $\mathcal{Q} = (q, C, \mathcal{O})$ denote a reasoning problem, where q is the question, C is an optional context passage, and $\mathcal{O} = \{o_1, \dots, o_K\}$ is the candidate answer set for multiple-choice tasks or the output space for free-form tasks. A *reasoning trace* $\tau = (s_1, s_2, \dots, s_T)$ is an ordered sequence of natural-language reasoning steps, and a prediction function $\phi(\tau) \in \mathcal{O}$ extracts the final answer from the trace. Let $\mathcal{M}_\theta$ denote a language model with parameters θ .

2.2 Chain-of-Thought (CoT) Reasoning

In CoT prompting, the model generates a single reasoning trace autoregressively:

$$\begin{aligned} \tau^{\text{CoT}} &= (s_1, s_2, \dots, s_T) \\ s_t &\sim \mathcal{M}_\theta(\cdot \mid q, C, \mathcal{O}, s_{1:t-1}). \end{aligned} \quad (1)$$

Generation terminates when the model emits a terminal answer pattern, and the final prediction is $\hat{y}^{\text{CoT}} = \phi(\tau^{\text{CoT}})$.

Why linear generation can fail. Equation (1) defines exactly one path through the space of possible reasoning traces. Let $\mathcal{T}$ denote the set of valid traces for a problem. CoT explores one element $\tau^{\text{CoT}} \in \mathcal{T}$ with no mechanism to evaluate intermediate branches before committing, to backtrack when an early step is misleading, or to combine evidence from multiple partial trajectories. Alternatives, such as breadth-first search and depth-first search remove some of these constraints, but they are either computationally impractical or lack a principled admissibility guarantee.

2.3 A* Search over Reasoning Traces

We therefore search over the space of reasoning traces using an A*-style priority rule, but with a finite node budget and a multi-goal voting procedure for tractability.

State space. The search space is a tree $\mathcal{G} = (\mathcal{S}, \mathcal{E})$ where each node $n \in \mathcal{S}$ represents a partial reasoning state:

$$n = (\tau_n, d_n), \quad \tau_n = (s_1, \dots, s_{d_n}). \quad (2)$$

Here, τ_n is the partial trace stored at node n and $d_n = |\tau_n|$ is its depth. The root node n_0 has $\tau_{n_0} = \emptyset$ and $d_{n_0} = 0$. An edge $(n, n') \in \mathcal{E}$ exists when $\tau_{n'} = \tau_n \oplus s$ for some reasoning step s .

Successor generation. Given a node n with state (τ_n, d_n) , we ask the language model to propose N_c candidate next steps:

$$\{(s^{(j)}, c^{(j)})\}_{j=1}^{N_c} = \text{GS}(\mathcal{M}_\theta, q, C, \mathcal{O}, \tau_n, N_c). \quad (3)$$

GS refers to the step-generation function. Each candidate consists of step text $s^{(j)}$ and a self-reported confidence $c^{(j)} \in [0.0, 1.0]$. The candidate induces a successor node n'_j with trace $\tau_{n'_j} = \tau_n \oplus s^{(j)}$ and depth $d_{n'_j} = d_n + 1$.

Goal condition. A node is a goal node if its last step contains a terminal answer pattern:

$$\text{ISGOAL}(n) = \mathbb{1}[\exists p \in \mathcal{P} : p \subset s_{d_n}], \quad (4)$$

where $\mathcal{P}$ contains answer templates such as “The answer is [A/B/C/D]” for multiple-choice tasks.

Motivating Example	Query: “Is shrimp scampi definitely free of plastic?”	Gold: FALSE
✗ Chain-of-Thought <i>Linear generation</i> <i>Commits early to a chronological preparation schema and overlooks the central biological premise.</i> Step 1. Shrimp harvesting: Shrimp are caught or farmed; while microplastics may exist in seawater, there is no conclusive evidence that shrimp contain significant plastic. Step 2. Processing: Cleaning and preparation do not necessarily involve plastic contamination. Step 3. Cooking: Cooking itself does not introduce plastic. Step 4. Ingredients: No direct evidence suggests serious contamination from ingredients. Step 5. Oversight: Regulatory standards make contamination unlikely. Answer. YES (Incorrect) ✓ A* Search <i>Heuristic-guided</i> <i>Prioritizes the most relevant premise and avoids a distracting step-by-step preparation narrative.</i> Step 1. Seafood, including shrimp, often contains microplastics in tissues and digestive systems. Step 2. Standard shrimp scampi preparation does not remove microplastics already present in the shrimp. Answer. FALSE (Correct)		

Figure 1: Comparison of reasoning trajectories on a StrategyQA example. CoT follows a plausible but misdirected chronological schema, while A* prioritizes the key biological fact—microplastic presence in shrimp—and reaches the correct conclusion with fewer, more relevant steps.

Path cost. The cumulative cost of a node trades off reasoning length and uncertainty:

$$g(n) = \sum_{t=1}^{d_n} [1 + (1 - c_t)] = d_n + \sum_{t=1}^{d_n} (1 - c_t). \quad (5)$$

Each step contributes a base cost of 1, and lower-confidence steps incur an additional penalty. This makes shorter, higher-confidence traces cheaper than longer or more uncertain ones.

2.4 Heuristic Design and Admissibility Guarantee

The heuristic estimates the remaining cost from a partial trace to a valid goal. We combine a depth-based term with a coverage term:

$$h(n) = h_{\text{depth}}(n) + h_{\text{coverage}}(n), \quad (6)$$

where

$$h_{\text{depth}}(n) = \omega_d \cdot \max(0, H_{\min} - d_n), \quad (7)$$

and

$$h_{\text{coverage}}(n) = \omega_c \cdot (1 - E(n)). \quad (8)$$

Here $H_{\min}$ is a task-level lower bound on the number of reasoning hops required from the root (a hyperparameter which is fixed per dataset), $E(n) \in [0, 1]$ is the fraction of target entities already covered by the trace at node n , and $\omega_d, \omega_c > 0$ are

heuristic weights. All of these are hyperparameters that are pre-determined for a particular dataset. Let

$$c_{\min} = 1 + (1 - c_{\max_conf}) \quad (9)$$

denote the minimum possible cost of a single transition under Eq. (5).

The confidence and coverage terms should be interpreted as practical heuristics for search control rather than calibrated uncertainty estimates or exact measures of evidential completeness.

Assumption 1 (Minimum-hop lower bound). *Any valid goal trace requires at least $H_{\min}$ reasoning steps from the root. Equivalently, for any non-goal node with depth $d_n < H_{\min}$, at least $H_{\min} - d_n$ further transitions are required before reaching a goal.*

Assumption 2 (Non-goal continuation). *Any non-goal node with depth $d_n \geq H_{\min}$ still requires at least one additional transition before reaching a goal.*

Theorem 1 (Conservative admissibility). *Under Assumptions 1 and 2, if $\omega_d + \omega_c \leq c_{\min}$, then the heuristic in Eq. (6) is admissible for all non-goal nodes; that is, $h(n) \leq h^*(n)$, where $h^*(n)$ is the true minimum remaining path cost.*

Proof. We consider three cases.

Case 1: root node ($d_n = 0$). By Assumption 1, any valid goal trace requires at least $H_{\min}$ future

transitions. Since each future transition costs at least $c_{\min}$, we have

$$h^*(n) \geq H_{\min} c_{\min}.$$

At the root, $E(n) \in [0, 1]$, so

$$h(n) = \omega_d H_{\min} + \omega_c (1 - E(n)) \leq \omega_d H_{\min} + \omega_c.$$

Using $\omega_d + \omega_c \leq c_{\min}$ and $\omega_d \leq c_{\min}$, we obtain

$$\begin{aligned} \omega_d H_{\min} + \omega_c &= (H_{\min} - 1)\omega_d + (\omega_d + \omega_c) \\ &\leq (H_{\min} - 1)c_{\min} + c_{\min} = H_{\min} c_{\min} \leq h^*(n). \end{aligned} \quad (10)$$

Case 2: intermediate node ($0 < d_n < H_{\min}$). Let $m = H_{\min} - d_n \geq 1$. By Assumption 1, at least m further transitions are still required, so $h^*(n) \geq m c_{\min}$. Also,

$$\begin{aligned} h(n) &= \omega_d m + \omega_c (1 - E(n)) \leq \omega_d m + \omega_c \\ &= (m - 1)\omega_d + (\omega_d + \omega_c) \leq (m - 1)c_{\min} + c_{\min} \\ &= m c_{\min} \leq h^*(n). \end{aligned} \quad (11)$$

Case 3: deep non-goal node ($d_n \geq H_{\min}$). By Assumption 2, at least one further transition is required, so $h^*(n) \geq c_{\min}$. Since $h_{\text{depth}}(n) = 0$ in this case,

$$\begin{aligned} h(n) &= \omega_c (1 - E(n)) \leq \omega_c \leq \omega_d + \omega_c \leq c_{\min} \\ &\leq h^*(n). \end{aligned} \quad (12)$$

Thus $h(n) \leq h^*(n)$ in all cases, proving admissibility. $\square$

For the main experiments, we keep the search hyperparameters fixed across datasets to avoid over-tuning the search procedure itself. In particular, the reported results should be interpreted as evaluating a single budgeted A*-style configuration rather than an extensively tuned search system. Dataset wise A* search hyperparameter values are shown in Table 6.

Algorithm 1 summarizes the overall inference procedure.

2.5 Routing Strategies

Given the observed complementarity between CoT and A*, we define routing functions $\mathcal{R} : \mathcal{Q} \rightarrow \{\text{CoT}, A^*\}$ that select the strategy on a per-instance basis.

Semantic entropy (Kossen et al., 2024). We compute the entropy of the answer distribution obtained from self-consistency sampling under CoT. High-entropy instances are routed to A*, while low-entropy instances keep the CoT answer.

Feature-based random forest. We extract hand-crafted features from the problem text, including question length, logical connectives, negation markers, and question-type indicators, and train a random forest classifier to predict which reasoning strategy will succeed.

LLM-as-critic. When CoT and A* disagree, we invoke a stronger critic model $\mathcal{M}_{\text{critic}}$ (DeepSeek-R1-Distill-Qwen-14B). The critic receives both traces and returns a verdict over which trace better supports its answer:

$$\text{verdict} = \mathcal{M}_{\text{critic}}(\text{Trace}_{\text{CoT}}, \text{Trace}_{A^*}, q, C, \mathcal{O}) \quad (13)$$

When the two methods already agree, we use the shared answer directly and do not call the critic.

3 Results

We evaluate on three benchmarks using two models: i) LLaMA-3.1-8B and ii) Qwen-0.5B and DeepSeek-R1-Distill-Qwen-14B as the critic. StrategyQA (Geva et al., 2021) tests multi-step commonsense reasoning with boolean answers. LogicQA (Liu et al., 2020) tests formal logical reasoning with four-way multiple choice. HotpotQA (Yang et al., 2018) tests multi-hop question answering with free-form answers. We report the exact accuracies from the current experimental pipeline, and the oracle corresponds to instance-wise selection of the correct answer between CoT and A*.

3.1 Quantitative Analysis

Main quantitative takeaway. Table 1 shows that no single reasoning strategy dominates across all settings. A* is substantially stronger than CoT on StrategyQA and HotpotQA, while CoT has a slight edge on LogicQA. Yet on every dataset the oracle sits far above either individual strategy, which confirms strong complementarity rather than simple superiority of one method over the other.

Oracle gaps. Relative to the stronger single method, the oracle improves by +9.79 points on StrategyQA, +13.51 on LogicQA, and +14.10 on HotpotQA. These are large margins. They indicate that even when one strategy wins on average, the losing strategy still solves many instances that the winner misses.

Router performance. Refer to Table 2. Routing closes only part of this gap. On StrategyQA, the

Dataset	Base methods on LLaMA-3.1-8B				Base methods on Qwen-0.5B				Best Router	Oracle
	CoT	A*	SC	ToT	CoT	A*	SC	ToT	RF	
StrategyQA	64.52	<u>74.80</u>	71.22	72.24	53.60	53.40	<u>54.21</u>	52.24	76.25	84.59
LogicQA	45.78	44.85	46.10	<u>46.21</u>	<u>24.00</u>	20.20	22.10	21.19	48.20	59.29
HotpotQA	76.80	<u>80.10</u>	78.21	78.45	8.40	<u>30.00</u>	12.21	22.17	81.25	94.20

Table 1: Main results across the three benchmarks. SC denotes Self-Consistency and ToT denotes Tree-of-Thought. RF denotes the rule-based random forest router, which is the best-performing router on all three datasets. Router results are evaluated only on LLaMA-3.1-8B. Bold marks the best non-oracle result in each row. All router improvements over CoT are statistically significant under McNemar’s test ($p < 0.005$). Best single-method results are underlined.

Router	StrategyQA	LogicQA	HotpotQA
Semantic Entropy Router	74.37	47.47	81.30
LLM-as-Critic Router	74.19	47.16	80.21
LLM-as-Judge Router	73.28	47.21	80.87
Rule-based Random Forest	76.25	48.20	81.25

Table 2: Router performance on top of LLaMA-3.1-8B. Bold marks the best router on each dataset. All router improvements over CoT are statistically significant under McNemar’s test ($p < 0.005$). Differences between the LLM-as-Judge and LLM-as-Critic routers arise only from prompt design; the corresponding prompts are provided in Appendix Section 2.

feature-based random forest reaches 76.25%, improving by +1.45 over the stronger single method (A*). On LogicQA, the same router reaches 48.20%, a +2.42 gain over CoT. On HotpotQA, semantic entropy performs best at 81.30%, a +1.20 gain over A*. These gains are real but much smaller than the oracle margin, which means the central difficulty is identifying *when* a given strategy should be used.

Scalable versus feature-engineered routing. The strongest absolute router on StrategyQA and LogicQA is the feature-based random forest, but it relies on hand-crafted, dataset-sensitive features. The more portable strategies are semantic entropy and LLM-as-critic. Among those, semantic entropy is the strongest overall on HotpotQA and competitive on LogicQA, while the critic is close but not consistently best.

What the critic helps with. The LLM-as-critic router only runs on disputed cases. On StrategyQA, the disagreement set has size 576 out of 2,290 examples, which is about 25% of the evaluation set. This makes the critic computationally attractive. However, its aggregate gains are modest: it improves over CoT on LogicQA and slightly over A* on HotpotQA, but on StrategyQA it remains below A*. This suggests that adjudicating between traces

is itself difficult, especially when A* traces are less fluent than CoT traces even when they are logically better.

Dataset	Both correct	CoT only	A* only	Both wrong
StrategyQA ($n=2290$)	1361 (59.4%)	328 (14.3%)	248 (10.8%)	353 (15.4%)
LogicQA ($n=651$)	180 (27.6%)	118 (18.1%)	88 (13.5%)	265 (40.7%)
HotpotQA ($n=500$)	298 (59.6%)	86 (17.2%)	87 (17.4%)	29 (5.8%)

Table 3: Instance-level complementarity decomposition. “CoT only” means CoT is correct and A* is wrong; “A* only” means the reverse.

Instance-level decomposition. Table 3 makes the complementarity concrete. HotpotQA is especially striking: only 5.8% of instances are missed by both strategies, while 34.6% lie in a regime where exactly one strategy is correct. StrategyQA also exhibits sizable disagreement. LogicQA is harder overall, as shown by the 40.7% “both wrong” mass, but even there each strategy recovers failures of the other.

3.2 Qualitative Analysis

To understand *why* CoT and A* differ, we manually inspected representative disagreement cases across StrategyQA, LogicQA, and HotpotQA. Rather than treating qualitative analysis as a collection of isolated examples, we organize these cases into recurring *reasoning regimes*. The resulting picture is consistent with the quantitative results: A* is not uniformly better than CoT, but helps in problem settings where reasoning requires exploration over multiple plausible partial trajectories, while CoT remains competitive or preferable when the solution path is naturally linear and can be executed reliably in a single forward pass. Detailed examples for each category are provided in Appendix A.

When A* helps. Across all three datasets, A* is most useful when the reasoning problem has a *branching structure*: multiple partial hypotheses must be considered, compared, or eliminated

before the correct conclusion becomes clear. We observe three especially common patterns.

First, A* helps in *non-linear evidence aggregation*, where the decisive evidence is distributed across several facts that do not naturally compose into a single fluent chain. In such cases, CoT often commits early to one locally plausible narrative and then continues refining that narrative even when it is globally suboptimal. Search mitigates this by allowing alternative branches to remain alive long enough for the more informative path to surface.

Second, A* is advantageous in *hypothesis elimination* problems, especially in multiple-choice logical reasoning. Here the difficulty is often not deriving one direct proof, but ruling out superficially plausible options that match the wording while failing structurally. A* can allocate reasoning budget across alternatives and compare them more explicitly, whereas CoT may lock onto the first option whose surface form appears compatible with the prompt.

Third, A* is helpful in *multi-hop composition*, particularly on HotpotQA-style questions where the answer depends on traversing intermediate entities or relations. In these examples, search acts less like a source of extra knowledge and more like a mechanism for preserving multiple partial chains until the relevant bridge entity is found.

Why these gains arise. The common thread in the above cases is that A* provides a way to defer commitment. CoT generates a single left-to-right trajectory, which is efficient when that trajectory is correct, but brittle when early steps are only partially informative or mildly misleading. In contrast, A* imposes structure over the reasoning process: it treats intermediate thoughts as expandable states, and this makes it possible to revisit the search space when the first promising line of reasoning turns out not to be the best one. Qualitatively, the advantage is therefore *better allocation of reasoning effort across competing partial solutions*.

When CoT remains preferable. The same analysis also makes clear that search is not always beneficial. CoT often performs better when the task is effectively *single-path reasoning*: the answer follows from direct factual recall, a short deterministic comparison, or a coherent argumentative trajectory that does not benefit from branching. In these cases, A* can degrade performance by introducing unnecessary exploration. We observe two frequent regimes where this happens.

The first is *direct factual reasoning*, where the problem reduces to retrieving a few stable facts and combining them straightforwardly. CoT often succeeds because a single fluent chain is enough, whereas A* may overcomplicate the problem by searching over semantically adjacent but irrelevant distinctions.

The second is *globally coherent argumentation*, especially in some LogicQA instances. These problems reward maintaining a stable abstract frame over the entire reasoning process. CoT can preserve such a frame naturally through a single explanation, while A* may fragment the reasoning into locally plausible steps that do not combine into the strongest global argument.

Characteristic failure modes of A*. Our manual inspection also reveals that A* has systematic failure modes rather than random errors. The most prominent one is *overthinking simple problems*: when the answer is accessible via direct recall or a short inference, search may create unnecessary distinctions and drift away from the obvious solution. A second failure mode is *spurious associative expansion*, where one branch of the search picks up a weak semantic connection and the search process continues elaborating that connection despite its limited causal relevance to the final answer. A third is *budget waste through repetition or near-duplicate expansions*, where search spends steps rephrasing or revisiting the same intermediate fact instead of exploring genuinely new possibilities.

These failure modes are important because they show that the cost of search is not merely computational. Search changes the *shape* of reasoning, and this can hurt when the original problem does not require exploration in the first place.

Dataset-level interpretation. The qualitative patterns also help explain the dataset-level behavior in Table 1. StrategyQA and HotpotQA contain many examples where evidence must be combined across partially independent clues or hops, and these are precisely the settings where A* more often improves over CoT. By contrast, LogicQA contains many items where success depends on preserving the right logical frame or evaluating compact argument structures. In such settings, branching can help on some elimination-style questions but can also interfere with the kind of disciplined global coherence that CoT sometimes maintains better. This matches the empirical observation that

A* yields larger gains on StrategyQA and HotpotQA than on LogicQA.

Takeaway. The CoT–A* tradeoff is best understood in terms of *reasoning topology*: A* helps when the solution requires exploring multiple plausible paths, while CoT remains stronger when the reasoning is naturally linear and coherent. Thus, A* is most valuable not as a universal replacement for CoT, but as a better inductive bias for inherently search-like questions.

3.3 Human evaluation

We conduct a human evaluation of reasoning-trace quality using the criteria in §3.3. Two post-graduate annotators with experience in LLM reasoning research each labeled 100 instances from StrategyQA and LogicQA. Across datasets and metrics, Cohen’s κ ranges from 0.59 to 0.68, indicating fair agreement.

Annotation criteria. *Coverage* measures whether a trace reaches the required inferential path and supports the final conclusion. We use a 3-point scale: 1 if the reasoning is mostly incorrect and does not reach the conclusion, 2 if it is mostly correct but still fails to reach the conclusion, and 3 if both the reasoning and conclusion are correct.

Relevance measures how focused the trace is on concepts actually needed to answer the question. We score it coarsely on a 3-point scale based on the proportion of relevant steps, where relevant steps remain on-topic and avoid introducing unnecessary concepts.

Relevance is less diagnostic. In StrategyQA, CoT remains more relevant than A* when both are wrong (2.82 vs. 1.55), while in LogicQA both methods stay moderately relevant even when incorrect (2.76 for CoT, 2.53 for A*). Thus, staying on-topic is not enough: a trace may remain relevant yet still miss the decisive inferential steps. Overall, the annotations indicate that the main difference between CoT and A* on disagreement cases lies in *which method better covers the decisive reasoning steps*, while relevance is only weakly associated with correctness.

Interpretation. Refer to Table 5. Across both datasets, the method that is correct on a disagreement case consistently receives a much higher *coverage* score than the method that is wrong. On StrategyQA, when CoT is correct and A* is wrong, CoT achieves average coverage 3.00 versus 1.20 for A*; when A* is correct and CoT is wrong, A*

Dataset	# disagreement cases	Annotation sample
StrategyQA	587	100
LogicQA	332	100

Table 4: Size of the disagreement pool used for human-style analysis. We consider only instances where CoT and A* produce different final predictions. In single-label QA, the *both-right* bucket is therefore expected to be empty, which is confirmed in both datasets.

scores 2.71 versus 1.63 for CoT. The same pattern holds on LogicQA (3.00 vs. 1.86 for CoT-correct cases, and 2.77 vs. 1.88 for A*-correct cases). This suggests that disagreement cases are driven mainly by *coverage of the required reasoning path*, not by superficial stylistic differences.

4 Discussion

The real challenge is routing, not raw complementarity. The oracle gaps remain much larger than the realized router gains. Even the best router is 8.34 points below the oracle on StrategyQA and 12.90 points below it on HotpotQA. The main bottleneck is therefore strategy selection, not simply adding search as an alternative decoder.

Why routing remains hard. The critic router is especially informative. Although it can inspect both candidate traces and should therefore have an advantage over entropy-based routing, its gains are only modest. A likely reason is that CoT traces are often more fluent, while A* traces are more fragmented or branch-local. As a result, the critic may favor the trace that appears better written rather than the one with stronger reasoning. This points to trace normalization (rewriting) or compression as a natural next step.

When search is worth paying for. Our qualitative analysis suggests a fairly clear boundary. Search helps when reasoning requires non-linear evidence aggregation, hypothesis branching, or multi-hop chaining. CoT is often better when the answer follows from direct factual recall or a single coherent argumentative path. Routing can thus be viewed as predicting the reasoning structure a question requires.

What the theorem contributes. The theorem provides formal support for the search objective by showing that, under the stated assumptions, the heuristic is a conservative lower bound on path cost.

Dataset	Correctness group	n	CoT Cov.	CoT Rel.	A* Cov.	A* Rel.
StrategyQA	both wrong	11	1.00	2.82	1.09	1.55
	CoT right, A* wrong	51	3.00	2.57	1.20	2.16
	CoT wrong, A* right	38	1.63	2.47	2.71	2.00
LogicQA	both wrong	38	1.79	2.76	1.89	2.53
	CoT right, A* wrong	36	3.00	2.81	1.86	2.47
	CoT wrong, A* right	26	1.88	2.81	2.77	2.42

Table 5: Human-style annotation on 100 sampled disagreement cases per dataset. “Cov.” denotes reasoning coverage (1–3), and “Rel.” denotes relevance (1–3). Coverage measures whether the reasoning substantially reaches a correct conclusion; relevance measures the proportion of steps that stay on-topic and avoid unnecessary concepts.

In this way, it justifies the A*-style admissibility that is guaranteed.

5 Related Work

Chain-of-thought and structured reasoning. CoT prompting (Wei et al., 2022) and self-consistency (Wang et al., 2023) established the usefulness of explicit intermediate reasoning. Tree-of-Thoughts (Yao et al., 2024) and Graph-of-Thoughts (Besta et al., 2024) extended this line by exploring richer reasoning structures. Our work is closest in spirit to Tree-of-Thoughts, but differs in using an A*-style priority rule and in explicitly analyzing complementarity with standard CoT.

Search-augmented LLM reasoning. Reasoning via planning or search has been explored with Monte Carlo Tree Search (Hao et al., 2023), beam search (Xie et al., 2024), and other search variants. Our contribution in this space is narrower but more specific: we define a cost function over natural-language reasoning traces, pair it with a conservative admissibility result for the heuristic, and study the resulting search procedure as a complementary strategy rather than as a universal replacement for CoT.

LLM-based evaluation and verification. LLM-as-a-judge methods (Zheng et al., 2023) and step-level verification approaches such as process supervision (Lightman et al., 2023) motivate our LLM-as-critic router. We use a stronger model to adjudicate between traces only when CoT and A* disagree, which turns verification into a targeted routing problem rather than a full re-reasoning pass on every example.

Reasoning failures and compositionality gaps. Prior work has documented systematic reasoning failures even in strong language models. Press et al. (Press et al., 2023) showed that models often fail when individually solvable subproblems must

be composed. Huang et al. (Huang et al., 2023) argued that models do not reliably self-correct without external feedback. These findings motivate our treatment of CoT and A* as complementary external strategies rather than expecting a single trace generator to repair itself.

Routing and adaptive computation. Routing has also been studied at the architectural level, especially in mixture-of-experts systems (Shazeer et al., 2017). Our setting is different: we route between *reasoning algorithms* at inference time rather than between model components during a forward pass.

6 Conclusion

We benchmarked standard chain-of-thought (CoT) reasoning against a budgeted A*-style search procedure over reasoning traces on StrategyQA, LogicQA, and HotpotQA, evaluating both standalone performance and their complementarity through oracle and routing analyses. The main finding is that CoT and A* are strongly complementary: across benchmarks, each method often succeeds where the other fails, yielding large oracle gains and showing that non-linear search can add value beyond linear reasoning. At the same time, current routers recover only a limited fraction of this potential, and the strongest practical router remains partly feature-engineered, suggesting that the main challenge is not whether search can help, but when it should be invoked. Overall, our results position A* not as a replacement for CoT, but as a complementary reasoning strategy that is most useful when linear traces become brittle or prematurely committed. A natural direction for future work is therefore adaptive inference-time strategy selection: designing stronger and more portable routers, integrating CoT and search more tightly within a single reasoning process, and extending the comparison to larger models, broader benchmarks, and more compute-aware search procedures.

Limitations

While our study provides a useful first comparison of CoT and budgeted A*-style reasoning, there are several natural boundaries to its current scope. Our experiments focus on three representative benchmarks and one main model family, which already support the central complementarity claim, though broader evaluation across additional tasks and models would further clarify how widely the observed patterns extend. In addition, the strongest router on two benchmarks uses feature-engineered signals, which is informative for analysis but may be less portable than lighter-weight entropy-based or critic-based alternatives. Finally, the heuristic relies on self-reported confidence and entity-coverage signals, which serve as practical proxies for prioritizing the search frontier but could likely be improved with stronger calibration or more faithful relevance estimates.

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A Qualitative Examples

This appendix provides the detailed examples corresponding to the qualitative conclusions summarized in Section 3.2.

Hyperparameter	Symbol	StrategyQA	LogicQA	HotpotQA
Heuristic form	Eq. (6)–(8)	$h_{\text{depth}} + h_{\text{coverage}}$	$\frac{1}{2}(h_{\text{remaining}} + h_{\text{logic}})$	$h_{\text{depth}} + h_{\text{coverage}}$
Minimum hop lower bound	$H_{\min}$	2	–	2
Depth weight	ω_d	0.3	–	0.3
Coverage weight	ω_c	0.3	–	0.3
Logic-complexity weight	λ	–	0.12	–
Confidence cap	$c_{\max_conf}$	0.99	0.99	0.99
Branching factor	N_c	2	2 (n_candidates)	3
Maximum search depth	$D_{\max}$	8	5	8
Maximum explored nodes	B_{nodes}	15	12	30
Minimum goal depth	–	2	–	2
Vote threshold	–	2 matching answers	2 goal nodes	2 matching answers
Branch temperatures	–	[0.2, 0.6]	0.7 (normal), 0.05 (forced)	[0.15, 0.45, 0.75]
Forced conclusion trigger	–	$d \geq D_{\max} - 1$	$d \geq 4$	$d \geq D_{\max} - 1$
Early goal guard	–	if $d < 2$: conf. ≥ 0.75	–	if $d < 2$: conf. ≥ 0.85 , grounding ≥ 0.5
Ungrounded-answer penalty	–	–	–	+0.8 path cost if grounding < 0.3

Table 6: A* search hyperparameters extracted from the dataset-specific implementations. StrategyQA and HotpotQA use the depth-plus-coverage heuristic from Section 2.4, whereas LogicQA uses a distinct heuristic based on remaining depth and logical complexity.

A.1 Patterns of A* Success and CoT Failure

We identify three recurring settings where A* succeeds while CoT fails.

Group 1: Non-linear evidence aggregation.

These are questions where the answer requires combining several facts that do not naturally form a single left-to-right narrative.

StrategyQA — Non-linear Evidence Aggregation

Q: *Is shrimp scampi definitely free of plastic?* (Gold: NO)

CoT (× predicts YES): Proceeds through five sequential steps analyzing harvesting, processing, cooking, ingredients, and regulations. Each step independently concludes no significant plastic risk. The linear structure prevents cross-referencing marine biology evidence with the food preparation chain.

A* (✓ predicts NO): Step 1 directly identifies microplastic bioaccumulation in shrimp tissue. Step 2 notes that cooking does not remove microplastics. The heuristic guides the search toward the most relevant evidence node, avoiding the exhaustive but misleading step-by-step analysis.

LogicQA — Non-linear Evidence Aggregation

Q: *Based on the above statement [spatial arrangement of communities around Civic Park], which of the following can be derived?* (Gold: A)

CoT (× predicts D): Analyzes spatial relationships sequentially, accumulating errors in relative positioning. By the fourth step, the accumulated imprecision leads to selecting an incorrect spatial relationship.

A* (✓ predicts A): Explores multiple spatial constraint combinations in parallel. The search finds a consistent assignment by simultaneously satisfying the “southwest of” and “southeast of” constraints, correctly deriving that Civic Park is north of the administrative service area.

Group 2: Hypothesis elimination. A* is also effective when the problem naturally decomposes into evaluating or eliminating alternatives.

LogicQA — Hypothesis Elimination

Q: Which of the following is most similar to the argument: “All landscape rooms can see the landscape, but Li Wenbing’s family can’t see the landscape. Therefore, Li Wenbing’s family is not a landscape room.” (Gold: D)

CoT (× predicts C): Identifies the logical structure (modus tollens) but then incorrectly maps it to option C (a modus ponens structure) due to surface similarity.

A* (✓ predicts D): Separately evaluates options through different branches. One branch identifies the negative inference pattern in the premise and correctly maps it to option D (“Anyone who meets conditions can apply; Sun Wen did not apply; therefore Sun Wen did not meet conditions”), which preserves the modus tollens structure.

Group 3: Multi-hop question answering. HotpotQA highlights the value of search when the answer depends on chaining facts across sources.

HotpotQA — Multi-hop Chaining

Q: What government position was held by the woman who portrayed Corliss Archer in the film Kiss and Tell? (Gold: Chief of Protocol)

CoT (× predicts “Ambassador”): Correctly identifies Shirley Temple as the actress but then retrieves an incorrect government position from parametric memory, confusing ambassadorial roles with her protocol role.

A*: Explores multiple evidence paths for Shirley Temple’s government career. While the trace is imperfect, the tree search structure allows it to consider multiple candidate positions before committing.

A.2 Patterns of CoT Success and A* Failure

We identify two common settings where CoT remains better than search.

Group 1: Direct factual reasoning. When the answer follows directly from familiar facts, search can introduce unnecessary branching and semantic drift.

StrategyQA — Direct Factual Reasoning

Q: Is a Boeing 737 cost covered by Wonder Woman (2017 film) box office receipts? (Gold: YES)

CoT (✓ predicts YES): Directly retrieves the production budget (~\$150M), box office (~\$822M), and Boeing 737 cost (~\$100M), concluding that the receipts exceed the aircraft cost.

A* (× predicts NO): The first search step focuses on whether box office receipts are “associated with” aircraft purchases, misreading the question as one about financial transactions rather than magnitude comparison.

StrategyQA — Direct Factual Reasoning

Q: Is average number of peas in a pod enough commas for a billion? (Gold: YES)

CoT (✓ predicts YES): Correctly identifies that a billion (1,000,000,000) requires three commas and that a pea pod usually contains 7–9 peas, so the answer is yes.

A* (× predicts NO): Incorrectly states that a billion has 12 zeros and becomes confused by the scale comparison.

Group 2: Fluent world knowledge and argumentation. When the problem is best solved by a smooth, coherent narrative, CoT often remains more reliable than fragmented search steps.

LogicQA — Fluent Argumentation

Q: Which of the following is the best argument against the claim that rich Chinese are transferring property abroad? (Gold: B)

CoT (✓ predicts B): Systematically compares the answer options and identifies the strongest counterargument.

A* ($\times$ predicts A): The first search step identifies option A as relevant, and the heuristic keeps the search near that branch without sufficiently comparing the full option set.

A.3 A* Failure Modes on Simple Cases

We observe three recurrent failure modes for A* on examples that CoT solves easily.

Failure mode 1: Overthinking simple questions.

StrategyQA — Overthinking

Q: *Does Disney have an ice princess?*
(Gold: YES)

A* ($\times$ predicts NO): Step 1 correctly identifies Elsa from *Frozen* as having ice powers. Step 2 notes that she is called “The Ice Queen” in promotional material. The search then turns the easy recall problem into an unnecessary semantic distinction between “queen” and “princess,” yielding the wrong answer.

Failure mode 2: Spurious correlations in search.

StrategyQA — Spurious Correlation

Q: *Is Menthol associated with Thanksgiving?* (Gold: NO)

A* ($\times$ predicts YES): Step 1 associates menthol with cough drops used in cold weather. Step 2 associates Thanksgiving with cold weather in North America. Step 3 turns these weak associations into a spurious chain, effectively reasoning menthol $\rightarrow$ cold weather $\rightarrow$ Thanksgiving.

Failure mode 3: Step repetition.

StrategyQA — Step Repetition

Q: *Will the Albany in Georgia reach a hundred thousand occupants before the one in New York?* (Gold: NO)

A* ($\times$ predicts YES): Step 2 states the population of Albany, New York ($\sim 98,000$). Step 3 largely repeats the same fact. The repetition wastes node budget that should have been used to compare growth trajectories.

Algorithm 1 Budgeted A* Search over Reasoning Traces

Require: Problem $\mathcal{Q} = (q, C, \mathcal{O})$, model $\mathcal{M}_\theta$, depth limit $D_{\max}$, node budget B , candidates N_c

```

1: Initialize root  $n_0 \leftarrow (\emptyset, 0)$ ,  $g(n_0) \leftarrow 0$ ,
    $h(n_0) \leftarrow h(n_0)$  from Eq. (6)
2: OPEN  $\leftarrow \{n_0\}$  (priority queue on  $f(n) = g(n) + h(n)$ )
3: VISITED  $\leftarrow \emptyset$ , GOALS  $\leftarrow \emptyset$ , explored  $\leftarrow 0$ 
4: while OPEN  $\neq \emptyset$  and explored  $< B$  do
5:    $n \leftarrow \text{OPEN.pop\_min}()$ 
6:   if sig( $n$ )  $\in$  VISITED then
7:     continue
8:   end if
9:   VISITED  $\leftarrow \text{VISITED} \cup \{\text{sig}(n)\}$ 
10:  explored  $\leftarrow$  explored + 1
11:  if ISGOAL( $n$ ) then
12:    GOALS  $\leftarrow \text{GOALS} \cup \{n\}$ 
13:    continue
14:  end if
15:  if  $d_n \geq D_{\max}$  then
16:    continue
17:  end if
18:   $\{(s^{(j)}, c^{(j)})\}_{j=1}^{N_c} \leftarrow \text{GENERATE\_STEPS}(\mathcal{M}_\theta, q, C, \mathcal{O}, \tau_n, N_c)$ 
19:  for each  $(s^{(j)}, c^{(j)})$  do
20:     $n' \leftarrow (\tau_n \oplus s^{(j)}, d_n + 1)$ 
21:     $g(n') \leftarrow g(n) + 1 + (1 - c^{(j)})$ 
22:    OPEN.push( $n', g(n') + h(n')$ )
23:  end for
24: end while
25: return weighted vote over GOALS

```

B FAQs

Q1. Why are the gains on HotpotQA much larger than on StrategyQA? HotpotQA explicitly rewards multi-hop evidence aggregation from supporting passages, where A* benefits from exploring multiple reasoning paths and prioritizing entity coverage. StrategyQA is often solvable with a coherent linear explanation, so CoT remains competitive and the two methods appear more complementary than hierarchical.

Q2. LogicQA shows A* underperforming CoT. Why include it at all? Because it gives a more honest picture of the method. LogicQA stresses formal sequential reasoning, where branching can introduce plausible but globally inconsistent paths, so reporting it avoids overstating A*’s benefits.

LLM-as-Judge Purpose. Compare two candidate reasoning trajectories and decide which one solves the query more <i>faithfully</i> and <i>correctly</i>. Inputs. • Context • Query • Path A reasoning trace • Path B reasoning trace Evaluation criteria. • Faithfulness: uses only the provided context and avoids invented constraints • Logical validity: each step follows causally and coherently from prior steps • Grounded conclusion: the final answer directly resolves the query from the established reasoning Required output. The model first reasons internally about both traces, then returns only the winning path in constrained XML: <winner>A</winner> or <winner>B</winner>	LLM-as-Critic Purpose. Diagnose the weaknesses of two reasoning traces and determine which trace is better overall. Inputs. • Question • Answer options • Trace A with predicted answer • Trace B with predicted answer Evaluation criteria. • whether the reasoning actually entails the predicted answer • whether there are logical fallacies or unsupported jumps • whether any claims are hallucinated or ungrounded Required output. The model returns a structured JSON critique with flaws for both traces and the preferred trace: <pre>{ "trace_A_flaws": "...", "trace_B_flaws": "...", "better_trace": "A/B" }</pre>
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Figure 2: LLM-based routing prompts. The *judge* prompt is verdict-oriented: it compares two reasoning paths under faithfulness and logical soundness criteria and outputs only the winner. The *critic* prompt is diagnosis-oriented: it identifies flaws in each trace before selecting the better one in structured JSON form.

Q3. Do the large oracle gaps mean the heuristic is weak, or is A* fundamentally limited? Both factors matter. The heuristic is imperfect, especially on LogicQA where it is not formally admissible, but many failures also come from the underlying model’s reasoning limits rather than the search procedure itself.

Q4. Why not report router results if the oracle suggests strong routing potential? The router outputs from development were not logged under a sufficiently uniform evaluation protocol. We therefore report oracle bounds as motivation, but leave router benchmarking to future work rather than present inconsistent numbers.

Q5. How reliable are self-reported confidence scores as search costs? They are not calibrated probabilities, and we do not treat them as such. They serve as pragmatic ranking signals that bias search toward shorter and more confident traces, while the clamping avoids pathological costs.

Q6. How much more compute does A* require than CoT? We report node-count proxies rather than exact token usage: 4.78, 2.72, and 9.57 explored nodes on StrategyQA, LogicQA, and Hot-

potQA. This suggests A* is tractable, but precise token-level accounting remains future work.

Q7. How should readers interpret the admissibility result? It is intentionally narrow. The proof applies only to the coverage-based heuristic on StrategyQA and HotpotQA, and does not imply that the full budgeted system is globally optimal.

Q8. Can you quantify how often the qualitative patterns occur? Not yet in a systematic way. The qualitative examples are diagnostic rather than exhaustive, and a proper frequency estimate would require dedicated annotation or automated categorization.

Q9. Why emphasize Qwen-0.5B in the paired comparisons? Because those runs preserve the cleanest matched CoT/A* evaluation setup. We also report LLaMA results, but the Qwen comparisons provide the strongest basis for controlled paired analysis.

Q10. Why use the cost function $g(n) = d_n + \sum_t (1 - c_t)$? It gives a simple additive tradeoff between reasoning length and uncertainty. This is a design choice rather than a theorem, but it worked

reasonably across datasets without causing either trivial collapse or uncontrolled exploration.

Q11. What are the main hurdles for deployment? The main issues are latency, compute cost, and the lack of a strong router. A* needs multiple serial model calls per query, so deployment only makes sense in settings where higher reasoning reliability justifies that overhead.

Q12. Would simple ensemble voting capture the same complementarity? Only partially. Self-consistency can recover some diversity, but it does not explicitly guide exploration toward promising reasoning states the way heuristic search does.

Q13. Why not report breakdowns by question type or error category? That is a limitation of the current study. We provide qualitative patterns and complementarity counts, but a systematic error taxonomy would require an additional annotation or classification effort.

Q14. Is the goal detection based on answer patterns brittle? Yes, somewhat. Pattern matching is fast and reproducible, but it can miss valid conclusions phrased differently or occasionally accept shallow conclusions too early.

Q15. Why stress that admissibility applies only to non-goal expansion priority? Because otherwise readers might wrongly infer that the full algorithm is optimal. The guarantee only concerns the ordering induced by the heuristic under stated assumptions; the practical system still uses budgets, voting, and early termination.